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GitHub Link	https://github.com/kiduuuu/MLA0405-Deep-learning

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME: Deep Learning **COURSE CODE: MLA0405**
SLOT : D

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10% **Duration: 90 minutes**
Assessment Tool Weightage: 30% **Mark : 25**

Code of Conduct:
I D.Chinna Moulali (192425278) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D.Chinna Moulali **Name of the student:** D.Chinna Moulali

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Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation **(2.5)**	Critical Analysis & Technical Evaluation **(2)**	Future Scope, Presentation & Documentation ***(1)***	**Total (10)**
Weightage	***20 %***	***25%***	***25%***	***20 %***	***10 %***	**100 %**
Q1						
Total						

QUESTIONS: 01 Total Marks:25

Select and review a recent research article related to the below **topic**, such as **Transfer Learning, DenseNet, PixelNet, RNN, Bidirectional RNN, Encoder–Decoder/Seq2Seq, BPTT, or LSTM**. Summarize the **problem addressed, dataset used, proposed deep learning model, methodology, performance metrics, major results, and key findings**, and critically comment on the **advantages, limitations, and possible future improvements** of the approach presented in the article.

Prepare a structured Article Review addressing the following sections:

1. Article Details
Provide the following information:
Title of the research article
Names of the authors
Journal/Conference name
Year of publication
DOI or publication link
Topic/Deep Learning technique addressed

2. Research Problem
What problem does the article attempt to solve?
Why is the problem important?
What are the limitations of existing approaches identified by the authors?
3. Objectives of the Study
What are the main objectives of the research?
What specific goals does the proposed model attempt to achieve?

4. Dataset Description
What dataset(s) are used?
What is the size of the dataset?
What type of data is used—images, text, time-series, video, etc.?
How is the dataset divided into training, validation, and testing sets?

5. Proposed Methodology
Explain the overall methodology adopted in the article.
Identify the deep learning architecture used.
Explain the important layers/components of the model.
Describe the data preprocessing techniques used.
Explain the training procedure and important parameters, wherever available.

6. Deep Learning Technique Used
Explain how the selected technique is applied in the research.
How is Transfer Learning used?
How does DenseNet perform feature reuse?
How is RNN/LSTM used to process sequential data?
How does a Bidirectional RNN utilize information from both directions?
How does an Encoder–Decoder/Seq2Seq architecture process input and output sequences?
How is BPTT used for training recurrent networks?

7. Model Architecture

Draw or reproduce a simplified architecture diagram of the proposed model and explain:
Input → Preprocessing → Feature Extraction/Encoder → Hidden Layers → Classification/Decoder → Output
Identify the role of the major components in the architecture.

8. Performance Evaluation

Identify the evaluation metrics used in the article, such as:
Accuracy, Precision, Recall, F1-Score , Loss , Mean Squared Error (MSE)
BLEU Score, where applicable
Report the major performance results obtained by the proposed model.

9. Results and Discussion

What are the major findings of the research?
How does the proposed model perform compared with existing methods?
Which model/technique produced the best results?
What factors contributed to the observed performance?

10. Advantages of the Proposed Model

Identify at least three advantages of the proposed approach.

11. Limitations of the Study

Identify the major limitations mentioned by the authors and any additional limitations you observe, such as: Small dataset,High computational requirements,Training time,Overfitting Limited generalization,Model complexity.

12. Critical Analysis

Provide your own analysis of the research article:
Is the selected deep learning architecture appropriate for the problem?
Is the dataset adequate?
Are the evaluation metrics appropriate?
Are the experimental results convincing?
Could another Unit IV technique provide better performance?

13. Future Scope

Suggest possible improvements or future research directions based on the limitations identified in the article.

14. Conclusion

Provide a brief conclusion summarizing:
The research problem
Proposed model
Major results
Your overall assessment of the article

15. References

Provide the complete citation of the selected research article and any additional references used in the review.

Article Review: EXPLAINABLE DEEP LEARNING FOR BREAST CANCER DETECTION

1. Article Details

Title

"An explainable AI-driven deep neural network for accurate breast cancer detection from histopathological and ultrasound images"

Authors

- Md. Romzan Alom
- Fahmid Al Farid
- Muhammad Aminur Rahaman
- Anichur Rahman
- Tanoy Debnath
- Abu Saleh Musa Miah
- Sarina Mansor

Journal

Scientific Reports, Volume 15, Article 17531, 2025.
Year of Publication:2025
The article was published on 20 May 2025.
DOI:10.1038/s41598-025-97718-5
Publication Link
Scientific Reports – Full Article

Topic / Deep Learning Techniques

The research mainly addresses:

- Transfer Learning
- DenseNet121
- Convolutional Neural Networks
- Image Classification
- Data Augmentation
- Explainable AI
- Grad-CAM

The proposed system is called Deep Neural Breast Cancer Detection (DNBCD). It uses DenseNet121 as the main backbone, customized CNN layers, transfer learning, and Grad-CAM for interpretability.

2. Research Problem

Breast cancer is a major healthcare problem where early and accurate diagnosis is extremely important.Traditional diagnosis depends heavily on expert examination of medical images. Such manual interpretation can be time-consuming and may vary between observers. The researchers identify two important challenges in existing deep learning systems:

1. Limited accuracy/generalization
2. Lack of interpretability

A highly accurate model may still be difficult for doctors to trust if it cannot explain why a particular image was classified as malignant or benign.The research therefore attempts to develop an automated breast-cancer classification system that is:

- Accurate
- Robust
- Data-efficient
- Explainable
- Suitable for medical-image analysis

The authors specifically use transfer learning because medical-image datasets are often limited compared with large general-purpose image datasets.

3. Objectives of the Study

The major objectives are:

Objective 1

Develop a deep learning model capable of classifying breast-cancer images accurately.

Objective 2

Use DenseNet121 with transfer learning to extract meaningful visual features.

Objective 3

Improve model robustness using:

- Image resizing
- Normalization
- Data augmentation
- Class weighting

Objective 4

Evaluate the proposed model on both:

- Histopathological images
- Ultrasound images

Objective 5

Improve interpretability using Grad-CAM so that important image regions contributing to predictions can be visualized.

Objective 6

Compare the proposed model with other CNN-based approaches such as:

- MobileNet
- ResNet50
- VGG19
- DenseNet
- Other state-of-the-art approaches

The overall goal is to develop a model that combines high classification performance with explainability.

4. Dataset Description

The researchers use two major datasets.

A. BreakHis-400x Dataset

The B-400x dataset is a subset of the BreakHis breast-cancer histopathological database.

It contains:

1,820 images

collected from:

82 patients

at 400× magnification.

The classes are:

Class	Number of Images
Benign	588
Malignant	1,232
Total	1,820

The dataset therefore has class imbalance because malignant images are more numerous than benign images.

B. BUSI Dataset

The second dataset is the:

Breast Ultrasound Images Dataset (BUSI)

It contains:

1,578 images

and contains three classes:

- Benign

- Malignant
- Normal

Therefore, the study evaluates the model on two different medical imaging modalities.

Dataset Types

Dataset	Data Type	Classes
B-400x	Histopathological images	Benign, Malignant
BUSI	Ultrasound images	Benign, Malignant, Normal

The use of two different datasets is valuable because it provides a broader evaluation of the proposed architecture.

Dataset Division

The authors use:

- 70% Training
- 10% Validation
- 20% Testing

For B-400x:

- Training = 1,274 images
- Testing = 364 images
- Validation = 182 images

The same 70/10/20 strategy is used for BUSI.

5. Proposed Methodology

The proposed DNBCD methodology consists of several stages.

Medical Image
↓
Image Preprocessing
↓
Data Augmentation
↓
DenseNet121
↓
Transfer Learning
↓
Additional CNN Layers
↓
Global Average Pooling
↓
Dense Layer
↓
Dropout
↓
Output Layer
↓
Classification
↓
Grad-CAM Explanation

The methodology begins with preprocessing the medical images.

The images are resized to:**128 × 128 × 3**

and pixel values are normalized to the range:**[0, 1]**

Data augmentation is then used to create variations of the training images.

Preprocessing
The paper uses several preprocessing techniques.
1. Resizing
Images are converted to:
128 × 128 pixels
with three RGB channels.
2. Normalization
Pixel values are divided by 255:
Normalized Pixel = Pixel Value / 255
This produces values between 0 and 1.
3. Data Augmentation
The researchers use transformations such as:

- Rotation up to 20°
- Width shifting
- Height shifting
- Shearing
- Zooming
- Horizontal flipping

These transformations increase the diversity of the training data and help reduce overfitting.
4. Class Weighting
Because the datasets are imbalanced, class weights are applied.
This gives greater importance to underrepresented classes during training.

6. Deep Learning Technique Used
Transfer Learning
Transfer learning is the central technique used in this research.
Instead of training a deep CNN completely from scratch, the researchers use a pre-trained DenseNet121 model.
The knowledge learned from a large image dataset is transferred to breast-cancer image classification.
Conceptually:

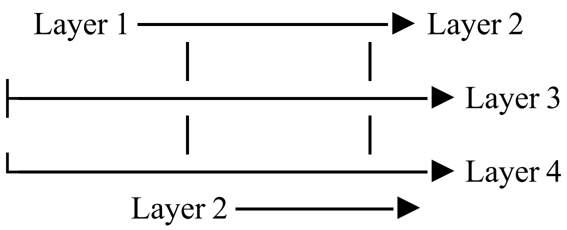
Large Image Dataset
↓
Pre-trained DenseNet121
↓
Learned Visual Features
↓
Breast Cancer Dataset
↓
Fine-tuning / Classification

This approach is useful when the target dataset is relatively small.

DenseNet Feature Reuse
DenseNet is designed around **dense connections**.
In a conventional CNN:

Layer 1 → Layer 2 → Layer 3 → Layer 4

In DenseNet:



Layer 3 ⟶

Each layer can access features generated by previous layers.
Therefore, features learned in earlier layers can be reused by later layers.
For medical images, early layers can learn:

- Edges
- Textures
- Shapes

while deeper layers learn:

- Tissue patterns
- Tumor structures
- Complex visual characteristics

Dense connections help preserve and reuse these features.

7. Model Architecture

The simplified DNBCD architecture is:

Input Medical Image
↓
Resize to 128 × 128
↓
Normalization
↓
Data Augmentation
↓
DenseNet121
Pre-trained Backbone
↓
Transfer Learning
↓
CNN Layers
↓
Global Average Pooling
↓
Dense Layer
↓
Dropout
↓
Dense Layer
↓
Sigmoid | Softmax

B-400x Dataset   BUSI Dataset

Benign/Malignant   Normal/Benign/Malignant

↓

Grad-CAM

↓

Visual Explanation

The study compares different transfer-learning backbones and selects the DenseNet121-based configuration as the proposed DNBCD model.

Major Components

DenseNet121
Acts as the primary feature extractor.

Global Average Pooling
Converts feature maps into a compact feature representation.

Dense Layers
Learn the relationship between extracted features and cancer classes.

Dropout
Reduces overfitting by randomly dropping neurons during training.

Sigmoid
Used for the binary B-400x classification.

Softmax
Used for the three-class BUSI classification.

Grad-CAM
Provides a visual explanation of which image regions influenced the prediction.

8. Performance Evaluation

The study evaluates the model using several metrics:

- Accuracy
- Precision
- Recall
- F1-score
- AUC
- Loss
- MAE
- RMSE

This is stronger than evaluating the model using accuracy alone.

B-400x Results

The proposed DNBCD model achieved:

Metric	Result
Accuracy	93.97%
Precision	97.47%
Recall	93.52%
F1-score	95.45%
AUC	99.24%
Loss	0.1607
MAE	0.0603
RMSE	0.2455

The model therefore achieved very strong classification performance on the B-400x dataset.

BUSI Results
On the BUSI dataset:

Metric	Result
Accuracy	89.87%
Precision	91.11%
Recall	89.87%
F1-score	90.00%
AUC	97.75%
Loss	0.4095
MAE	0.1392
RMSE	0.4639

The model achieved its strongest overall performance on the B-400x dataset, while still performing well on BUSI.

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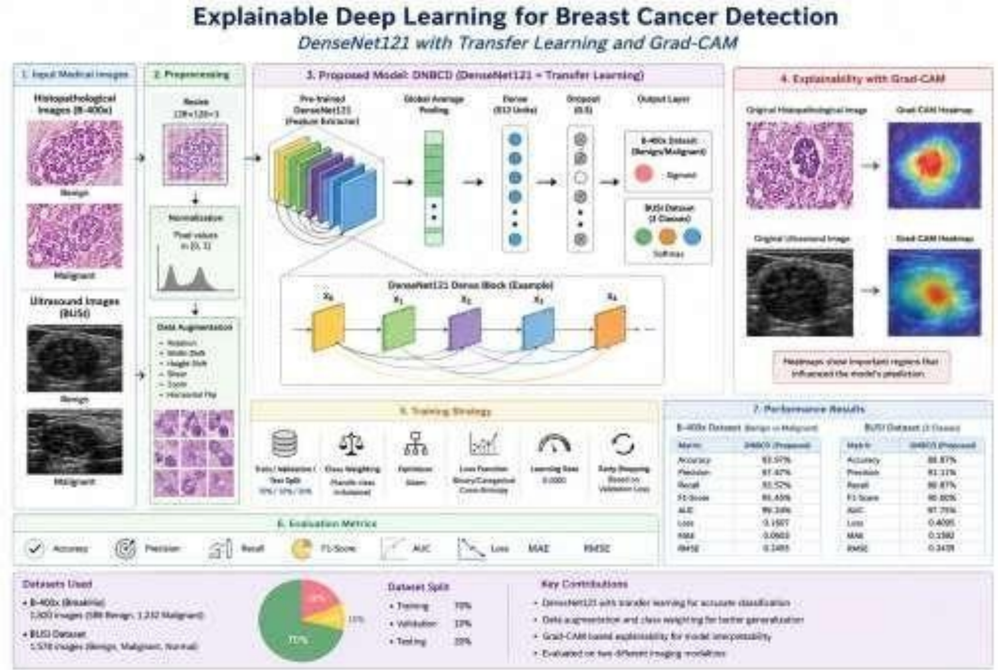
9. Results and Discussion
The results indicate that the proposed DNBCD model performs strongly on both datasets.
For B-400x:
Accuracy = 93.97%
F1-score = 95.45%
AUC = 99.24%
For BUSI:
Accuracy = 89.87%
F1-score = 90.00%
AUC = 97.75%
The researchers compared DNBCD with several existing approaches and reported that the proposed model achieved the highest overall performance in their comparison.
The study also compared different transfer-learning configurations involving models such as:

- DenseNet121
- MobileNet
- ResNet50
- VGG19

The DenseNet-based DNBCD configuration performed particularly well.



External Dataset Evaluation
An important strength of the study is that it did not rely only on its primary test sets.
The researchers additionally evaluated the model using BreakHis-100x as an external dataset.
One configuration achieved:
80.32% accuracy
when trained on B-400x and tested on B-100x.
When datasets were merged, some configurations achieved:

- 94.52%
- 98.47%
- 96.90%

This experiment provides additional evidence about the model's ability to generalize to related data, although the performance variation also demonstrates that cross-dataset generalization remains challenging.

10. Advantages of the Proposed Model

Advantage 1 – Transfer Learning
Using a pre-trained DenseNet121 reduces the need to train a very deep network from scratch.
This is especially useful for medical datasets where labeled data is limited.

Advantage 2 – Feature Reuse
DenseNet's dense connections allow earlier features to be reused by later layers.
This improves feature propagation and representation learning.

Advantage 3 – Strong Classification Performance
The model achieves:

- 93.97% accuracy on B-400x
- 89.87% accuracy on BUSI

and high AUC and F1 scores.

Advantage 4 – Explainability
Grad-CAM generates heatmaps showing the image regions that contributed to the model's prediction.
This is particularly valuable in healthcare because doctors can visually inspect the model's focus areas.

Advantage 5 – Multiple Imaging Modalities
The model is evaluated using both:

- Histopathological images
- Ultrasound images

This demonstrates its potential across different types of medical images.

11. Limitations of the Study
The authors identify several important limitations.

1. Dataset Diversity
The model mainly uses B-400x and BUSI datasets.
These may not represent the full diversity of images encountered in real clinical environments.

2. Image Quality Variation
Differences in:

- Lighting
- Resolution
- Image quality

- Imaging conditions

can reduce generalization.
The authors recommend using more diverse imaging conditions in future training.

3. Class Imbalance
Although class weighting is used, class imbalance remains a challenge.
Some classes have substantially fewer samples than others.
This can affect:

- Precision
- Recall
- F1-score

4. Computational Complexity
The proposed architecture is more computationally demanding than simpler CNN models.
The authors specifically note that computational time can make real-time clinical deployment difficult.

5. Potential Overfitting
Medical datasets are relatively small compared with the complexity of deep neural networks.
Although augmentation, dropout and transfer learning reduce this risk, overfitting remains a potential concern.

12. Critical Analysis
Is DenseNet121 appropriate?
Yes.
DenseNet121 is appropriate because the study deals with image classification and has relatively limited medical datasets.
Its feature-reuse mechanism helps the model learn rich representations without repeatedly relearning similar features.
Transfer learning further improves its suitability.

Is the Dataset Adequate?
The use of two datasets is a strength.
However, the total dataset size is still relatively small compared with large-scale computer-vision datasets.
More importantly, the datasets may not represent all:

- Hospitals
- Imaging devices
- Patient populations
- Demographic groups
- Image acquisition conditions

Therefore, external multi-center validation would strengthen the evidence.

Are the Evaluation Metrics Appropriate?
Yes.
Using only accuracy would not be sufficient for medical classification.
The study uses:

- Accuracy
- Precision
- Recall
- F1-score
- AUC

- Loss
- MAE
- RMSE

This provides a more comprehensive evaluation.
For clinical applications, sensitivity, specificity, AUC and calibration would also be particularly important.

Are the Results Convincing?
The results are promising because:

1. Two datasets are used.
2. Multiple models are compared.
3. Several evaluation metrics are reported.
4. Multiple-run averages and standard deviations are reported.
5. An external B-100x evaluation is included.
6. Grad-CAM provides interpretability.

The reported B-400x accuracy of 93.97% and AUC of 99.24% are strong results.
However, high performance on public datasets does not automatically prove clinical readiness.
External prospective clinical validation would still be required.

Could Another Deep Learning Technique Perform Better?
Potential alternatives include:
Vision Transformers
Vision Transformers can model long-range relationships between image regions and may perform well when sufficient data or pretraining is available.
EfficientNet
EfficientNet could provide a better balance between accuracy and computational cost.
Attention-Based CNNs
Attention mechanisms could allow the model to focus more strongly on clinically relevant regions.
Multimodal Learning
Combining:

Histopathology
+
Ultrasound
+
Clinical Information

could potentially provide more comprehensive diagnostic information.
Therefore, DenseNet121 is a strong choice, but it should be compared against newer architectures under the same experimental conditions.

13. Future Scope
Several improvements can be proposed.
1. Larger and More Diverse Datasets
Future research should use datasets collected from:

- Multiple hospitals
- Multiple countries
- Different imaging devices
- Different patient populations

This can improve generalization.

2. Multimodal Learning
Future systems could combine:

Ultrasound

+
Histopathology
+
MRI
+
Clinical Data

This could provide more comprehensive diagnostic information.
The authors specifically mention incorporating additional modalities such as MRI.

3. Advanced Attention Mechanisms
Attention mechanisms could be integrated with DenseNet to focus on clinically relevant areas. This could improve feature extraction and classification.

4. Better Class-Balancing Methods
Future work could investigate:
- Oversampling
- Synthetic data generation
- GAN-based augmentation
- Advanced ensemble methods

to improve performance on minority classes.

5. Model Compression
To reduce computational cost, future versions could use:
- Pruning
- Quantization
- Knowledge distillation
- Lightweight CNN architectures

This could make the model more suitable for real-time clinical applications.

6. Improved Explainability
Grad-CAM can be supplemented with:
- SHAP
- Integrated Gradients
- Attention visualization

to provide multiple forms of explanation.

7. Clinical Validation
Before real-world deployment, the model should undergo:
- External validation
- Prospective clinical testing
- Bias analysis
- Calibration analysis
- Robustness testing
- Human-AI comparison

This would help establish whether the system can reliably support clinical decision-making.

14. Conclusion

The reviewed research article presents the Deep Neural Breast Cancer Detection (DNBCD) system, which combines DenseNet121, transfer learning, customized CNN layers, preprocessing, data augmentation, class weighting and Grad-CAM explainability for breast cancer classification.The system was evaluated using the B-400x histopathological dataset and the BUSI ultrasound dataset.
The proposed model achieved:

- **93.97% accuracy and 99.24% AUC on B-400x**
- **89.87% accuracy and 97.75% AUC on BUSI**

These results demonstrate that DenseNet-based transfer learning can provide strong performance when working with medical images.The major strength of the research is that it does not focus only on classification accuracy. The addition of Grad-CAM makes the model more interpretable by showing the regions influencing predictions.

However, limitations remain, particularly regarding dataset diversity, class imbalance, computational complexity and generalization to real clinical environments.

Overall Assessment

The article is a strong and relevant example of transfer learning and DenseNet application in healthcare. The combination of high classification performance and explainability makes the approach practically interesting.However, before clinical deployment, the model should be validated using larger, diverse, multi-center datasets and evaluated under real-world clinical conditions.

15. References

Main Research Article

Alom, M. R., Farid, F. A., Rahaman, M. A., Rahman, A., Debnath, T., Miah, A. S. M., & Mansor, S. (2025). An explainable AI-driven deep neural network for accurate breast cancer detection from histopathological and ultrasound images. *Scientific Reports, 15*, 17531. https://doi.org/10.1038/s41598-025-97718-5.

Additional Reference

Ayana, G., Dese, K., Abagaro, A. M., et al. (2024). Multistage transfer learning for medical images. *Artificial Intelligence Review, 57*, 232. https://doi.org/10.1007/s10462-024-10855-7.

Dataset References

The reviewed study uses the publicly available BreakHis and BUSI datasets. The authors provide dataset-access information in the paper.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution

		industry standards			
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis